HQIV chart hypothesis: O-Maxwell scalar fields and the modified fluid vacuum source

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Abstract

This note is the **paper companion** to the **F2 typed bundle** in `Hqiv/Physics/HQIVFluidClosureScaffold.lean`: at a fixed chart point c on $\mathbb{R}^{1+3}$ (coordinates $\mathrm{Fin}\,4 \to \mathbb{R}$), the fluid closure inputs $(\varphi, \nabla\varphi, \dot{\delta}\theta', \nabla\dot{\delta}\theta')$ are identified with continuum scalars φ_F, $\dot{F}$, an electric-energy proxy E', and spatial gradients from `coordsGradientComponents`. **Not claimed:** a theorem that solutions of full MHD or Navier–Stokes coincide with O-Maxwell; only **hypothesis bookkeeping** aligned with Lean. See `AGENTS/FLUID_OMAXWELL_ROADMAP.md` § F2.

Index convention. $\nu = 0$ is time; $\nu \in \{1, 2, 3\}$ are spatial. The fluid uses $\mathrm{Fin}\,3 \to \mathbb{R}$ vectors; `spatialFin4` embeds $i \in \mathrm{Fin}\,3$ as $\nu = i + 1$.

Modified fluid vacuum momentum (Python / Lean). The spatial source $\mathbf{g}_{\mathrm{vac}}$ in `pyhqiv.fluid` matches `hqivVacuumMomentumSource3`:

$$\mathbf{g}_{\mathrm{vac}} = -\frac{\gamma}{6}\nabla(\varphi\,\dot{\delta}\theta') = -\frac{\gamma}{6}\big(\varphi\,\nabla\dot{\delta}\theta' + \dot{\delta}\theta'\,\nabla\varphi\big), \qquad \gamma = \gamma_{\mathrm{HQIV}} = \tfrac{2}{5}. \tag{1}$$

F2 chart hypothesis (Lean name: `OMaxwellFluidChartHypothesis`). At a basepoint c:

$$\varphi_{\mathrm{fluid}} = \varphi_F(c), \tag{2}$$

$$(\nabla\varphi)_{\mathrm{spatial}} = \big(\partial_\nu \varphi_F(c)\big)_{\nu=1,2,3} \qquad \text{via } \texttt{chartSpatialPhiGradient}, \tag{3}$$

$$\dot{\delta}\theta' = \delta\theta'(E') = \texttt{delta\_theta\_prime}(E') \qquad (\texttt{Hqiv.delta\textbackslash\_theta\textbackslash\_prime} \text{ in } \texttt{ModifiedMaxwell.lean}), \tag{4}$$

$$(\nabla\dot{\delta}\theta')_{\mathrm{spatial}} = \big(\partial_\nu \dot{F}(c)\big)_{\nu=1,2,3} \qquad \text{via } \texttt{chartSpatialDotGradient} \text{ — user-chosen scalar } \dot{F} \text{ on} \tag{5}$$

Under these equalities, `hqivVacuumMomentumSource3_of_OMaxwellFluidChartHypothesis` rewrites the vacuum source to depend only on $(\varphi_F, \dot{F}, c, E')$.

MHD (ideal / resistive) as composition. **Ideal MHD** couples fluid velocity $\mathbf{u}$, magnetic field $\mathbf{B}$, and electric field $\mathbf{E} = -\mathbf{u} \times \mathbf{B}$ (in suitable units). HQIV does not reprove MHD here; the **closure alignment** is: the same $(\varphi, \dot{\delta}\theta')$ slots that enter $\mathbf{g}_{\mathrm{vac}}$ and eddy viscosity (see `hqivEddyViscosity`) are the ones whose **chart gradients** are fixed by the F2 bundle when coupling to O-Maxwell. Resistive terms add $\eta\,\mathbf{J}$; current $\mathbf{J}$ is still only **hypothesis-mapped** to stress (see F2 table, "Plasma J" row in `FLUID_OMAXWELL_ROADMAP.md`).

Repository. `Hqiv/Physics/HQIVFluidClosureScaffold.lean`, `Hqiv/Physics/ContinuumOmaxwellClosure.lean`, `Hqiv/Physics/ModifiedMaxwell.lean`.